

INTELLIGENT REAR-COLLISION WARNING SYSTEM FOR CYCLISTS

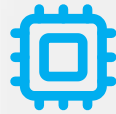
A SIXTH SENSE TO MAKE YOU SAFE ON THE ROAD.



24Ghz mm Wave Radar

Mounted under the seat facing rear to continuously detect approaching vehicles.

- 40 degrees of monitoring zone
- Split into following sectors
 1. Left = -7.5° to 0°
 2. Right = 0° to $+7.5^\circ$



Central Control Unit

ESP 32 WROOM 32E embedded on a custom designed PCB acting as the brain.



Feedback Units

Two ERM DC motors on left & right handlebars provide haptic feedback to the riders' arms via handlebar.



LEDs on top of haptic mount enclosures act as visual indicators.

SYSTEM FUNCTIONALITY

1

Detect Oncoming Threats in Real Time up to 35 m with a relative speed exceeding 10 kmph



2

Compute The Serial Data received from the Sensor and assess the Threat Level .

01
10

3

Alerts of both Haptic and Visual are executed to keep the rider aware.



4

Protect the rider from danger and made the ride a joyful exercise.

